

Lecture 8: Controllability & Observability

Angela Fontan

Dept. Decision and Control Systems
KTH Royal Institute of Technology
Sweden

English version of slides by ©Bo Wahlberg



Today's Ordlista

English	Svenska
state-space (form/representation)	tillstånds (form/beskrivning)
observability matrix	observability matrix
state vector	tillståndsvectorn
observability	observerbarhet
controllability	styrbarhet
identity matrix	enhetsmatrisen
range/image space	värderummet
null space	nollrummet
controllability matrix	styrbarhetsmatrisen
observability matrix	observerbarhetsmatrisen
controllable state	styrbar (tillståndsvektor)
observable state	observerbar (tillståndsvektor)



Outline

Recap: State-space representation

Exponential function

Controllability och Observability

Controllability

Observability

Minimal realization

In the previous lecture..

► **State-space representation:**

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), \quad x(0) = x_0 \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

- $x(t) \in \mathbb{R}^{n \times 1}$ state vector ($n \geq 2$)
- $u(t) \in \mathbb{R}^{m \times 1}$ input vector
- $y(t) \in \mathbb{R}^{p \times 1}$ output vector
- $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, $C \in \mathbb{R}^{p \times n}$, $D \in \mathbb{R}^{p \times m}$ (usually: $D = 0$)

Today:

- Exponential function
- Observability
- Controllability



Outline

Recap: State-space representation

Exponential function

Controllability och Observability

Controllability

Observability

Minimal realization

Exponential function

Goal: Solve (i.e., find the solution) of the state-space equations

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), \quad x(0) = x_0 \\ y(t) &= Cx(t)\end{aligned}$$

Trick*: Generalize the *integrating factor* ("integrerande faktor") $e^{-A\tau}$:

Since $\frac{d}{d\tau} e^{-A\tau} = e^{-A\tau}(-A)$

Then $(\dot{x} - Ax) = Bu \Rightarrow e^{-A\tau}(\dot{x} - Ax) = e^{-A\tau} Bu$

$$\Rightarrow \frac{d}{d\tau} (e^{-A\tau} x) = e^{-A\tau} Bu$$

$$\Rightarrow \underbrace{\int_0^t \frac{d}{d\tau} (e^{-A\tau} x) d\tau}_{e^{-At} x(t) - x(0)} = \int_0^t e^{-A\tau} Bu(\tau) d\tau$$


$$e^{-At}x(t) - x(0) = \int_0^t e^{-A\tau}Bu(\tau) \, d\tau$$
$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau) \, d\tau$$

6

Exponential function

How does one calculate e^{At} ?

Solve $\frac{d}{dt}e^{At} = Ae^{At}$, $e^{A \cdot 0} = I$.

1. Series expansion

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \dots$$

2. Laplace

$$e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$

Exponential function

3. Diagonalization

Suppose that A can be diagonalized:

$$A = TDT^{-1}, \quad D = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}, \quad (\text{eigenvalues})$$

Then $e^{At} = Te^{Dt}T^{-1}$, where

$$e^{Dt} = \begin{bmatrix} e^{\lambda_1 t} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n t} \end{bmatrix}$$

Proof via, for example, the series expansion, $A^k = TD^kT^{-1}$.

Observe: $e^{At} \xrightarrow{t \rightarrow \infty} 0$ if the eigenvalues of A , i.e., the poles of the system, lie in LPH

4. Via the Cayley-Hamilton's theorem.

Cayley-Hamilton's theorem

Each quadratic matrix satisfies its own characteristic equation: that is, if

$$\det(\lambda I - A) = \lambda^n + a_1 \lambda^{n-1} + \dots + a_{n-1} \lambda + a_n$$

then the following matrix equation holds

$$A^n + a_1 A^{n-1} + \dots + a_{n-1} A + a_n I = 0$$

Observation: if A^{-1} exists:

$$A^{-1} = \frac{-1}{a_n} (A^{n-1} + a_1 A^{n-2} + \dots + a_{n-1} I)$$

with $a_n = \det(-A)$


$$A^n = -a_1 A^{n-1} - \dots - a_{n-1} A - a_n I$$

Continue for $p \geq n$

Substitute in the series expansion

where $\{f_j(t)\}$ are linearly independent as functions of t .

Example

Assume that we have

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

We can then calculate

$$(sI - A)^{-1} = \frac{1}{s^2} \begin{bmatrix} s & 1 \\ 0 & s \end{bmatrix}$$

Which then helps us determine e^{At} as

$$e^{At} = \mathcal{L}^{-1} \left\{ \begin{bmatrix} \frac{1}{s} & \frac{1}{s^2} \\ 0 & \frac{1}{s} \end{bmatrix} \right\} = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

Note that $A^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = 0$, and therefore the series expansion for e^{At} is given exactly by $e^{At} = I + At$.



Outline

Recap: State-space representation

Exponential function

Controllability och Observability

- Controllability

- Observability

- Minimal realization

Example (8.13)

Assume that we have the system

$$\begin{cases} \dot{x} &= \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} u \\ y &= \begin{bmatrix} 2 & 1 & 0 \end{bmatrix} x \end{cases}$$

The system transfer function is given by

$$G(s) = C(sI - A)^{-1}B = \frac{2(s+2)(s+3)}{(s+1)(s+2)(s+3)} = \frac{2}{s+1}$$

Do we have the right number of states?

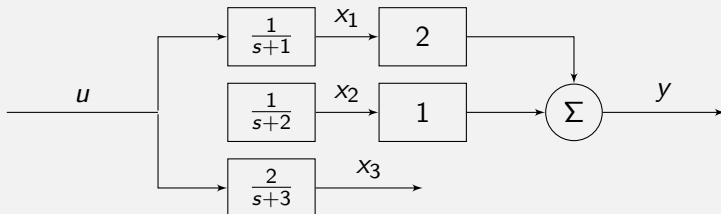
Controllability och Observability

Example (8.13 cont'd)

We find x_1 through the first row of the system of equations

$$\dot{x}_1 = -x_1 + u \Rightarrow X_1(s) = \frac{1}{s+1}U(s)$$

If we repeat the same procedure for x_2 , x_3 , and y , we can sketch the block diagram:

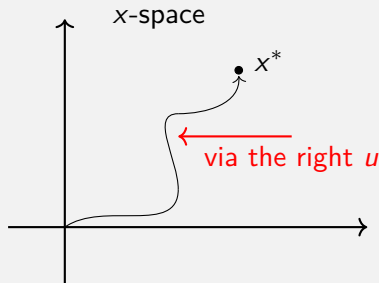


x_2 not affected by $u \Rightarrow$ Not controllable | x_3 not visible in $y \Rightarrow$ Not observable

Observe: $\dot{x}_2(t) = -2x_2(t) + 0u(t) \Rightarrow x_2(t) = e^{-2t}x_2(0) \Rightarrow$ does not depend on $u!$ $\Rightarrow$ Autonomous system: its evolution depends only on the initial state, not on any external input. In the block diagram, it is represented as a block without a "input arrow"

Definition (Controllability, Def. 8.3)

A state x^* is controllable if it is possible to control/steer the system from $x(0) = 0$ to $x(T) = x^*$ using some input u (in finite time T). A system is controllable if all the states are controllable.



$$x(t) = \int_0^t e^{A(t-\tau)} B u(\tau) d\tau \quad \text{where} \quad e^{At} = f_0(t)I + \dots + f_{n-1}(t)A^{n-1}$$

where $\gamma_k(t) = \int_0^t f_k(t - \tau)u(\tau)d\tau$

$$S = [B, AB, \dots A^{n-1}B]$$

[Resultat 8.9]

Suppose we have the system

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

We calculate the product

$$AB = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

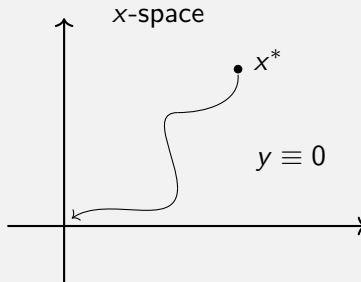
in order to be able to calculate the controllability matrix

$$\mathcal{S} = [B \ AB] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

whose determinant, $\det(\mathcal{S}) = -1 \neq 0$, tells us that the system is **controllable**.

Definition (Observability, Def. 8.4)

The state x^* is *unobservable* if the output signal is identically zero when the initial value is x^* and the input signal is identically zero.



Test of Observability

We have $u(t) \equiv 0$ and $x(0) = x^*$: $y(t) = Ce^{At}x^*$, $y^{(k)}(t) = CA^k e^{At}x^*$, $t \geq 0$.

In particular, $y \equiv 0$ if

$$y(0) = Cx^* = 0$$

$$\dot{y}(0) = CAx^* = 0$$

$$\vdots$$

$$y^{(n-1)}(0) = CA^{(n-1)}x^* = 0 \quad (\text{Why stop at } k = n - 1?)$$

That is, unobservable states x^* lie in the *null space* of **observability matrix**

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

The system is **observable** if $\mathcal{O}$ has full rank $= n$, i.e. if $\det(\mathcal{O}) \neq 0$

[Resultat 8.10]

Example (Test of observable states)

Suppose we have the system

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

We need to calculate the product $CA = \begin{bmatrix} 0 & 0 \end{bmatrix}$ to be able to derive the observability matrix

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Now, $\det \mathcal{O} = 0$, which means that the system **is not observable**.

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix}$$

$$\Rightarrow \mathcal{O} = \begin{bmatrix} 2 & 1 & 0 \\ -2 & -2 & 0 \\ 2 & 4 & 0 \end{bmatrix}, \quad \mathcal{O}_{x^*} = 0$$

$$\Rightarrow x^* = \begin{bmatrix} 0 \\ 0 \\ x_3 \end{bmatrix}$$

21

Definition (Minimal realization)

A system is a **minimal realization** if the input-output relationship cannot be described with fewer states.

Theorem (Resultat 8.12)

*A state-space representation is a **minimal realization** of an input-output relationship if and only if it is controllable and observable.*

Next time: Regulator design using state-space models